

3(a)(i)	Ttdd and TTdd ;	1
3(a)(ii)	13:3 ;	1
3(a)(iii)	epistasis ;	1
3(a)(iv)	any five from: 1 dominant allele / T / D, results in a functional protein ; 2 two recessive alleles / tt / dd, results in a non-functional protein ; 3 <i>T</i> could code for an enzyme ; 4 (so enzyme catalyses) production of, malvidin / pigment ; 5 <i>D</i> could code for an inhibitor ; 6 (inhibitor) binds to enzyme (coded for by) gene <i>T/t</i> ; 7 (so) reaction to produce, malvidin / pigment, does not occur ; 8 protein (coded by T / t) could be a transcription factor ; 9 transcription factor allow the production of enzyme (needed in the metabolic pathway) ; 10 protein (coded by <i>D/d</i>) could cause the inhibition of transcription factor ;	5

Question	Answer	Marks
3(b)	any four from: 1 located on X chromosome / sex-linked ; 2 <i>ref. to</i> recessive allele ; 3 non-functioning, factor VIII / protein or less, factor VIII / protein or no, factor VIII / protein ; 4 (so) blood does not clot quickly enough / excessive bleeding occurs (after an injury) ; 5 link the phenotype to the genotype ; e.g. $X^F X^F$ = normal female $X^F X^f$ = normal female $X^f X^f$ = affected female $X^F Y$ = normal male $X^f Y$ = affected male 6 AVP ; e.g. prevents activation of thrombin / fibrinogen not converted to fibrin / <i>ref. to</i> mostly males affected /different mutations cause a range of severity of haemophilia	4